

Statistical modeling

Module 12: Revisions

Marc-Olivier Boldi

Bachelor, GSEM

Autumn 2018

Directives for the exam

They are on the chamilo page.

- ▶ One final written exam at the end of the session (2h45).
- ▶ Restricted documentation: you are allowed to bring 12 A4 pages (6 sheets) prepared on your own. No writing style limitation.
- ▶ Exam type: open questions (maybe one or two multiple choices for fun).
- ▶ Includes: interpretations of results (e.g. of R outputs), analyses of situations (e.g. making recommendations in some situations), some calculations that can be performed with a simple calculator. If the exam is not testing your memory and your R programming skills, it is however clear that you are expected to be familiar with the functions treated in exercises and that you know the concepts treated in class. Otherwise, interpretations, analyses, and recommendations are useless. This is why the "limited authorized documentation" is large enough.

What we have studied 1

General and introduction:

- ▶ What is a model
- ▶ Dependence between 2 variables:
- ▶ Scatter plots
- ▶ Correlations: Pearson, Spearman, Kendall's
- ▶ Covariance

The regression equation and its assumptions

- ▶ Some theory: OLS and sum-of-squares, matrix notation, sum of squares decomposition
- ▶ Prediction equation
- ▶ Quality: R^2 , adjusted R^2 , plot observation versus fitted values

What we have studied 2

Categorical variables

- ▶ Dummy variables coding
- ▶ Interpretation of the coefficients, contrasts
- ▶ Interactions: cat-cat, num-num, cat-num, higher order interactions

Uncertainty:

- ▶ Standard error (standard deviation of the error)
- ▶ Prediction intervals
- ▶ Notions of model building

What we have studied 3

Inference:

- ▶ the fundamental concept, population, sample.
- ▶ F-Tests:
 - ▶ Model significance
 - ▶ Anova table
 - ▶ Model comparisons (and its link to the two previous ones)
 - ▶ Anova of type I, II and III
- ▶ Inference on the coefficients:
 - ▶ standard errors of the estimate, t-value and t-test
 - ▶ confidence intervals
 - ▶ test on linear combinations of coefficients

What we have studied 4

Regression assumptions:

- ▶ $e_i \sim N(0, \sigma^2)$ + independence
- ▶ Standardized residuals
- ▶ Diagnostics plots:
 - ▶ Residuals vs fitted values
 - ▶ QQ plots
 - ▶ Scale-location
 - ▶ Residuals versus leverage
- ▶ Notion of outliers and influential values.

Special topics:

- ▶ Moderation test
- ▶ Mediation test(s), total, direct and indirect effects.
- ▶ Association versus causality
- ▶ Analysis of collinearity, VIF.
- ▶ Variable selection: test based and AIC based approaches.

What we have studied 5

Logistic regression

- ▶ Binary outcome: graphical analysis
- ▶ Linear predictor, logit link function, predicted probability, predicted outcome.
- ▶ Interpretation of the coefficients (values on the linear scale, sign on the probability scale)
- ▶ Interactions
- ▶ Interpretation of predicted probability (equiv. prediction intervals)
- ▶ Estimation: Maximum likelihood and log-likelihood (theory)

Quality

- ▶ Confusion matrix,
- ▶ Accuracy, Sensitivity, specificity,
- ▶ ROC curve, AUC

What we have studied 6

Inference

- ▶ Comparison of models: LRT (incl. degree of freedom)
- ▶ Normal test on the coefficients
- ▶ Confidence intervals: normal based, others (profile likelihood)

Model assumptions

- ▶ Bernoulli distribution
- ▶ Deviance and Pearson residuals
- ▶ 4 diagnostics plots

GLM: concept and examples (type of outcomes, different distributions)

About the exam...

Possible calculations by hand (simple scientific calculator)

- ▶ Compute a prediction of a model given the coefficients (e.g. R output of a logistic regression), giving all the intermediate steps
- ▶ Compute a prediction interval
- ▶ Compute a confidence interval
- ▶ Identify an outlier given the number of data
- ▶ Compute an accuracy, sensitivity, specificity from a confusion matrix
- ▶ In any case: additions, subtractions, multiplications, divisions, power, exponential, logarithm, etc.

Other productions:

- ▶ Graphs of mediation and moderation in a given situation.
- ▶ Being able to complete a ROC curve given some results (e.g. specificities and sensitivities for several thresholds)
- ▶ Coding in dummy variables and interactions.
- ▶ Any other work on graphs showing you know what you are talking about.

About the exam...

Analysis:

- ▶ Be able to interpret any R output of the types that was used in the course, in the exercises, or that is similar to these.
- ▶ From some descriptive analysis (e.g. scatter plot) be able to anticipate some of the results of a further model, keeping the limits of these approach in mind.
- ▶ From R output, be able to make several steps of an analysis (e.g. a mediation analysis, a variable selection, etc.).

Discussions:

- ▶ Given a situation, data, etc. be able to propose and justify a type of analysis in full details (regression, logistic regression, which variables, why, moderation, mediation, etc.)
- ▶ Make proposition of improvements and what can be expected from it (e.g. change of variable, etc.)
- ▶ Others...

About the exam...

Type of exam questions

- ▶ Open questions
- ▶ Direct and without trap (or very very obvious ones, or not on purpose...)
- ▶ If you are not sure, write what you think is correct and explain.
- ▶ Technical questions (e.g. a calculation) are here to allow you to show that you know the concept and are able to do it. A small error in the numbers will not be severely sanctioned (but still do your best otherwise, little by little you lose points).